1 Equivalence relations

A relation on a set is a way of saying which elements are related. We begin with the definition and then prove that equivalence relations and partitions of a set are two descriptions of the same thing.

Definition 1.1
Let X be a set. A relation $\sim$ on X is an *equivalence relation* if it is reflexive ($x \sim x$ for all $x \in X$), symmetric ($x \sim y$ implies $y \sim x$) and transitive ($x \sim y$ and $y \sim z$ imply $x \sim z$).

Given an equivalence relation, each element determines the set of elements related to it, its *equivalence class* $[x] = \{y \in X \mid x \sim y\}$. The classes cover X, since $x \in [x]$, and two of them are either equal or disjoint.

Lemma 1.2
Let $\sim$ be an equivalence relation on X and let $x, y \in X$. Then $[x] = [y]$ if and only if $x \sim y$, and if $[x] \neq [y]$ then $[x] \cap [y] = \varnothing$.

Proof 1
Suppose $[x] = [y]$. Since $y \in [y] = [x]$ we have $x \sim y$. Conversely, suppose $x \sim y$ and let $z \in [y]$, so that $y \sim z$; by transitivity $x \sim z$, so $z \in [x]$. Hence $[y] \subseteq [x]$, and by symmetry $[x] \subseteq [y]$. Finally, if $z \in [x] \cap [y]$ then $x \sim z$ and $y \sim z$, so $x \sim y$ by symmetry and transitivity, and $[x] = [y]$. $\square$

Theorem 1.3
Let X be a set. For every equivalence relation $\sim$ on X the set $X/\!\sim = \{[x] \mid x \in X\}$ is a partition of X, and every partition of X arises in this way from exactly one equivalence relation.

Proof 2
That $X/\sim$ is a partition is the previous lemma together with $x \in [x]$. Conversely, let $\mathcal{U}$ be a partition of X and define $x \approx y$ if and only if x and y lie in the same block of $\mathcal{U}$.

- Each x lies in some block, so $x \approx x$.
- If x and y share a block then so do y and x.
- If x, y share a block U and y, z share a block V, then $y \in U \cap V$, and since blocks are disjoint or equal, $U = V$; so x, z share a block.

The equivalence classes of $\approx$ are exactly the blocks of $\mathcal{U}$:

$$\begin{aligned} [x]_{\approx} &= \{y \in X \mid x \approx y\} && \text{by definition} \\ &= \{y \in X \mid y \in U_x\} && \text{where } U_x \text{ is the block of } x \\ &= U_x. \end{aligned}$$

If two equivalence relations have the same classes they relate the same pairs, so the relation is unique. This completes the proof, which is long enough to show how a framed box is split across a page break: the frame is measured again for the first and the last part, and each part is drawn with its own picture. We add a few more lines of reasoning so that the box needs more room than the page has left. Suppose that $f : X \to Y$ is any function; then $x \sim y$ if and only if $f(x) = f(y)$ defines an equivalence relation whose classes are the non-empty fibres $f^{-1}[\{y\}]$, and the quotient map $q : X \to X/\sim$ is a surjection through which f factors uniquely as $f = \bar{f} \circ q$ with $\bar{f}$ injective. In particular $|X/\sim| = |f[X]|$. □

2 Modular arithmetic

Definition 2.1
Fix an integer $n > 0$. Integers a and b are *congruent modulo* n, written $a \equiv b \pmod{n}$, if n divides $a - b$.

Theorem 2.2
Congruence modulo n is an equivalence relation on $\mathbb{Z}$ with exactly n classes, $[0], [1], \ldots, [n-1]$.

Proof 3
Reflexivity and symmetry are clear since $n \mid 0$ and $n \mid -(a-b)$. If $n \mid a-b$ and $n \mid b-c$ then $n \mid (a-b)+(b-c) = a-c$. By the division theorem every a is $qn+r$ with $0 \leq r < n$, so $a \equiv r$, and two distinct remainders are not congruent since their difference has absolute value less than n. $\square$

Lemma 2.3
If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$ then $a+b \equiv a'+b' \pmod{n}$ and $ab \equiv a'b' \pmod{n}$.

Proof 4
We have $(a+b)-(a'+b') = (a-a')+(b-b')$ and $ab - a'b' = a(b-b')+b'(a-a')$, both multiples of n. $\square$